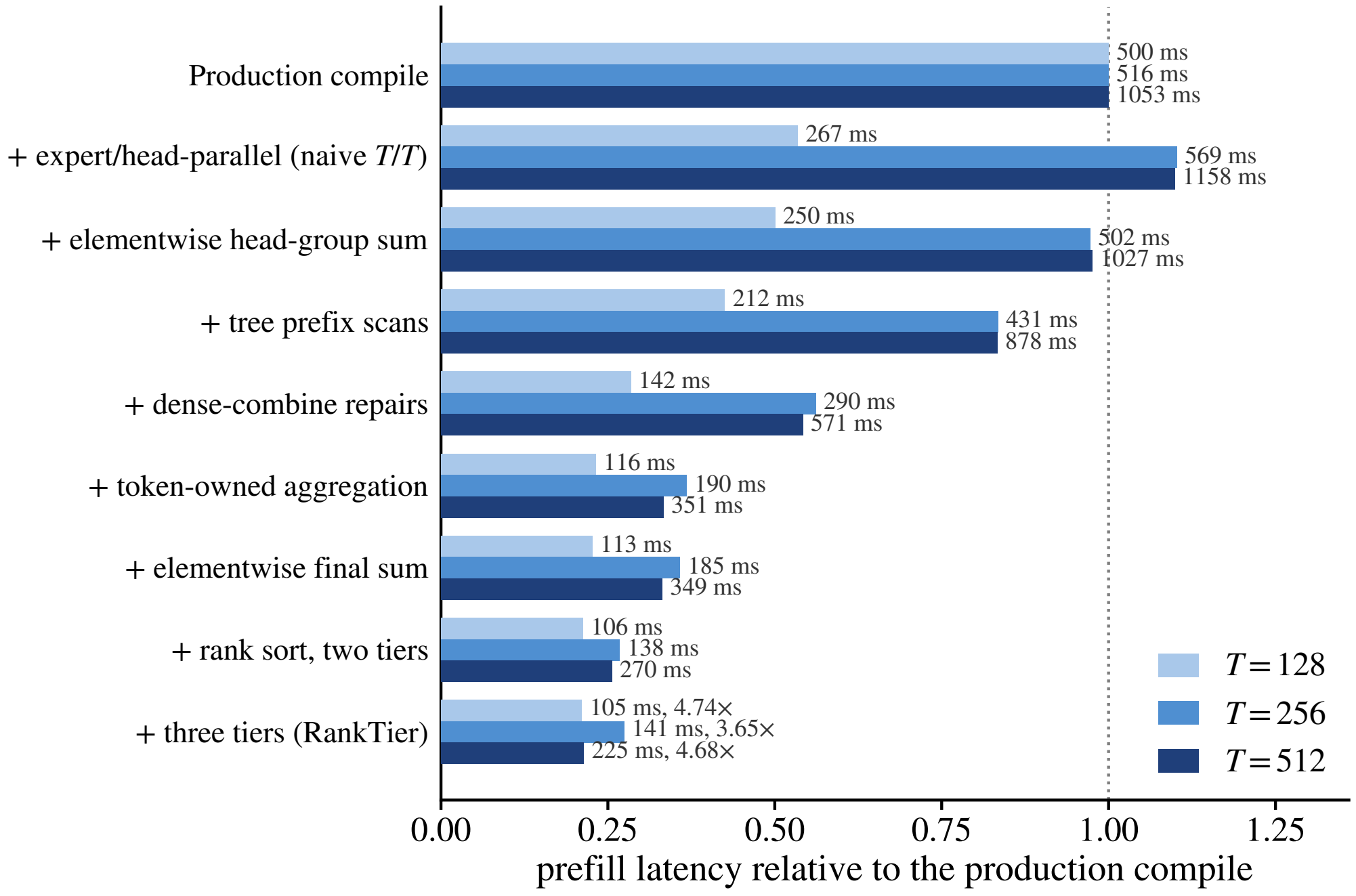
Production compile
+ expert/head-parallel (naive T/T)
+ elementwise head-group sum
+ tree prefix scans
+ dense-combine repairs
+ token-owned aggregation
+ elementwise final sum
+ rank sort, two tiers
+ three tiers (RankTier)
500 ms
516 ms
1053 ms
267 ms
569 ms
1158 ms
250 ms
502 ms
1027 ms
212 ms
431 ms
878 ms
142 ms
290 ms
571 ms
116 ms
190 ms
351 ms
113 ms
185 ms
349 ms
106 ms
138 ms
270 ms
105 ms, 4.74×
141 ms, 3.65×
225 ms, 4.68×
T = 128
T = 256
T = 512
0.00
0.25
0.50
0.75
1.00
1.25
prefill latency relative to the production compile